

technologies by building an ecosystem of right partnerships. It's time for them to reposition themselves in order to expand beyond traditional vehicle sales," he adds.

According to CyberMedia Research, a market research and consulting company, a CASE-led future requires automotive companies and technology giants to build on their traditional strengths. "Traditional automotive companies excel in hardware capabilities, and hardware-centric innovations. On the other hand, technology giants and tech start-ups bring cutting-edge capabilities, especially in areas such as compute, connectivity and computer vision, amongst others," says Prabh Ram, Head of Industry Intelligence Group at CyberMedia Research, adding that the success of traditional automotive companies as well as tech majors in a CASE paradigm will be invariably linked to how successful they are in building on their natural synergies.

Experts believe that tech giants will look at creating consumer differentiation and enduring value across the various tech layers in the vehicle. "When one looks at the overall complexity and the constraints that automotive manufacturing entails, as well as the high capital intensity and low margins, the appetite for tech companies to embrace full-scale automotive manufacturing will be lesser," says Ram.

Pacheco reckons that while technology has traditionally started at the upper automotive segments and cascaded to lower segments, this trend is different. "Even for high-end cars, tech wasn't being used for disruptive purposes. Today what we're seeing is disruptive innovation in terms of new revenue generation models, and a diversity of new services and features. It will change the industry in this decade," he says.

Both tech and auto companies are well aware of the challenges specific to the Indian market going forward. "The connected cars industry will need to overcome several challenges in its journey including the cost of connectivity, cybersecurity, and regulatory guidelines. Usage of these cars will generate large amounts of data that will draw specific usage patterns and

IN TODAY'S AUTOMOTIVE MARKET, CAR COMPANIES ARE NO LONGER MERELY ASSEMBLING MECHANICAL PARTS BUT DEVELOPING A COMPLICATED COMPUTER ON WHEELS

increase vulnerabilities," says Bansal.

He adds that vehicle manufacturers need to determine and evaluate these data sets thoroughly to make their products more effective and robust. Alongside, cybersecurity also requires a major focus. As more cars become connected, the efforts and resources required to manage connectivity and subscription will also increase significantly. "The OEMs and their partners will account for this transformation. Automotive companies must chalk out the business plans that can provide them with sufficient support, operational monitoring, and performance management. Apart from that, the sector would also require government support for the development of a conducive policy and regulatory environment," says Bansal.

With tech giants like Apple and Google becoming car manufacturers, where does one draw the line? "Frankly, nobody knows the answer. The automotive industry had a lot to offer to the world in terms of building at scale, cutting costs, managing inventories, etc. Now, tech companies have a lot to offer on the software side. But the lines are blurring," says Chaba.

That may just be good news for the consumer. **BT**

@PLidhoo, @nidhisigal